Darren Tang

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EDUCATION

Santa Clara University September 2025 - June 2027
Master of Science in Computer Science and Engineering
Relevant Coursework: Advanced Algorithms, Advanced Operating Systems, Artificial Intelligence, Object-Oriented Programming, Machine Learning, AI Governance and Cybersecurity, Mobile Application Development, Data Mining and Pattern Recognition.

University of Wisconsin-Madison September 2021 - May 2025
Bachelor of Science in Computer Science

EXPERIENCE

Incoming Graduate Researcher | SCU Video and Image Processing Lab, Santa Clara, CA January 2027 - Present
- Planned focus: applying deep learning to image and video analysis in computer vision research directed by Dr. Ying Liu.

AI Agent Development Intern | Veracyte, South San Francisco, CA June 2026 - September 2026
- Engineered a multi-agent system for CRM issue detection and reporting that cut weekly QC review from 25 to 15 hours.
- Created a Python sensitive data detector for LLM security and compliance, averaging 95% detection across thousands of samples.
- Developed an autonomous desktop AI agent using RAG across 10 enterprise sources to surface context during customer care calls.
- Conducted requirements gathering and functional testing for 8 automation initiatives across Customer Care and Laboratory teams.

Artificial Intelligence Analyst | DataAnnotation, Remote January 2026 - May 2026
- Built TypeScript and C++ testbeds evaluating LLM factuality, relevance, and reasoning using rubrics for RLHF and fine-tuning.
- Evaluated models, MCP integrations, and prompt variants through A/B tests to identify stronger-performing configurations.
- Ran LLM tool call and structured output evaluation in containerized Docker environments for reproducible and isolated testing.

Full Stack Software Engineer Intern | Sonic Engineering, Union City, CA June 2025 - September 2025
- Programmed a Java/SQL inventory service on Azure DevOps CI/CD that automated reorder checks, cutting review time by 50%.
- Collaborated with engineers and CNC operators in an Agile Scrum team to translate product requirements into validated features.
- Prototyped tool lookup, material tracking, and tool life monitoring features, iterating with CNC operators to improve workflows.

PROJECTS

MiniSpark | *C, pthreads, Linux* GitHub
- Built a parallel data processing engine in C modeled on Apache Spark that turns map, filter, join, and partitionBy calls into a DAG and computes each partition once its inputs are ready.
- Wrote a thread pool sized to the CPU core count with a shared work queue guarded by mutexes and condition variables, plus a logging thread that times each task, and passed all 21 concurrency tests.

MusicBloom | *React, TypeScript, FastAPI, SQLAlchemy, Docker* GitHub
- Built a React and TypeScript frontend on a versioned FastAPI REST API, with service and repository layers that store playback sessions, queues, and listening history through SQLAlchemy and Alembic migrations.
- Kept backend test coverage at 100% with pytest and automated linting, type checks, and tests in Azure Pipelines CI.

Tempest | *Python, scikit-learn, FastAPI, React* GitHub
- Trained models that predict whether SPY will close higher one and five days out, comparing logistic regression, random forest, and gradient boosting with time ordered validation across 5,400 trading days.
- Built 26 technical indicators like RSI, MACD, and Bollinger Bands without future data leaking into training, and served forecasts with explanations through FastAPI to a React dashboard.

HONORS AND AWARDS

- Lumen Scholarship Fund, $40,000 merit-based scholarship August 2021 - May 2025
- Dean's List, University of Wisconsin-Madison May 2024
- President's Volunteer Service Award, Gold, 250 volunteer hours Nov 2019 – Aug 2020
- Academic All American Award, National Speech and Debate Association June 2020

SKILLS

Certification: GitHub Foundations
Languages: Java, Python, TypeScript, JavaScript, C, C++, SQL, HTML/CSS
Technologies: React, Node.js, FastAPI, Spring Boot, PostgreSQL, REST APIs, pandas, NumPy, PyTorch, LangChain, RAG, MCP, Docker, Azure DevOps, CI/CD, Git, Linux, Salesforce Agentforce, Microsoft Copilot Studio, Claude Code, Cursor, Codex